Routing games

Braess’s paradox, in the room

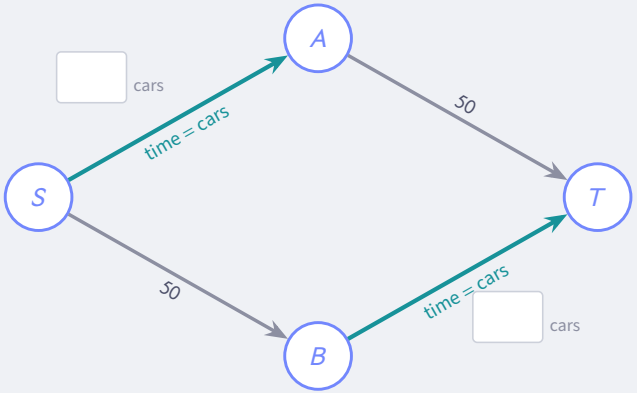
Phase 1
Two routes. Find where the class settles.

Phase 2
Add a road. Watch it get worse.

Then
Write it down properly.

Drivers in the room:

Date:



Top route

____ + 50 = ____

Bottom route

____ + 50 = ____

Stand on your route. Re-choose if the other one is quicker. Stop when nobody moves.

Round	On top	On bottom	Top time	Bottom time	Who moves?
1					
2					
3					
4					

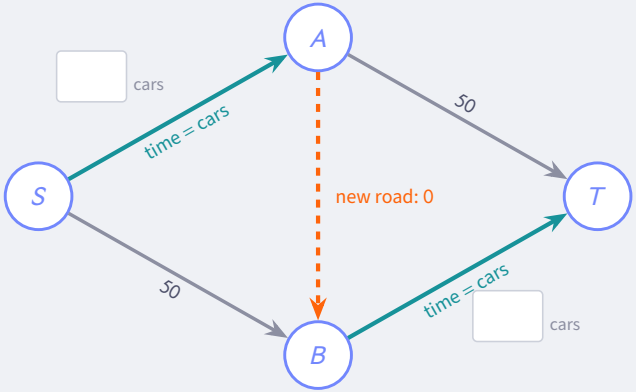
Where we settled: ____ on top, ____ on bottom.

If one driver switches from top to bottom, their time goes from ____ to ____.

So this is a _____ flow.

A flow is a Nash flow when:

Is it also the best the class could do?



$S \to A \to T$

$S \to B \to T$

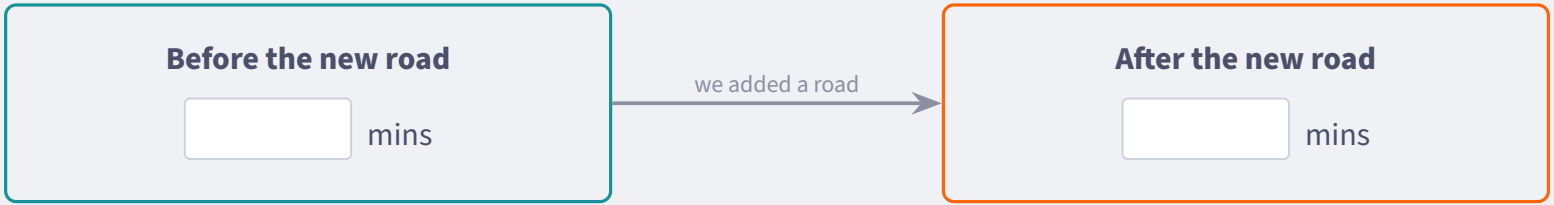
$S \to A \to B \to T$

Re-choose again. The new road is free, so who would not take it?

$S \to A$: ____ cars = ____ mins + $A \to B$: 0 mins + $B \to T$: ____ cars = ____ mins

Everyone's time: ____ mins

Nobody wants to leave: going back to $S \to A \to T$ would cost ____ + 50 = ____.



One more car joins a small road that already carries x cars.

It costs *me* ____ minutes.

It costs *everyone else* ____ minutes.

I pay the first and ignore the second. The road's total time is

$x\,c(x) =$ ____

and one more car really adds $\frac{\mathrm{d}}{\mathrm{d}x}\left(x\,c(x)\right) =$ ____

A routing game has a flow f on each route, and a cost

$$C(f) = \sum_e f_e \, c_e(f_e)$$

Our network, *before* the new road:

Our network, *after* the new road:

A planner who could route us all could ignore the new road and send half of us each way.

Left alone: the Nash flow $\tilde{f}$, and each driver's time:

Planned: the optimal flow f^*, and each driver's time:

$$\text{Price of anarchy} = \frac{C(\tilde{f})}{C(f^*)} = \frac{\square}{\square} = \square$$

What we just did in the room is **Question 1** on the Routing Games page, worth 25 marks.

(a) What a Nash flow is here. [3]

(b) The Nash flow and each driver's time. [5]

(c) Adding $A \to B$: show all 40 on $S \to A \to B \to T$ is a Nash flow. [7]

(d) Compare the times, and explain it through the congestion each driver ignores. [7]

(e) Compute the price of anarchy. [3]

vknight.org/gt/topics/routing-games.html